

Beyond Context-Decoupled Propagation: Context-Coupled Multimodal Graph Learning for Recommendation

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Abstract

Multimodal graph recommenders enrich user-item interactions with visual and textual evidence, but commonly process modal information and relation types in separate branches. Each branch therefore updates representations within its own propagation context, determined by the evidence, graph relations, normalization, and intermediate node representations available to it, while the resulting representations are combined only after propagation. This context-decoupled paradigm prevents heterogeneous information and relations from jointly shaping representation evolution and repeatedly executes sparse graph operations while maintaining multiple representation streams. We propose CC-MGR_{EC}, an instantiation of a context-coupled paradigm for multimodal recommendation. Rather than introducing additional evidence, CC-MGR_{EC} reorganizes existing evidence within one graph-learning context. It first grounds content-induced item relations in training behavior, then couples the calibrated semantic relations with observed interactions under shared normalization and propagates a single stream of node representations. This organization allows heterogeneous information to jointly shape the same evolving node representations and collaborative and semantic transitions to compose during propagation, while reducing branch-wise graph passes and intermediate-representation maintenance. Theoretically, we show that the resulting shared representation stream and joint operator realize compositions across both information sources and relation types and we establish their computational advantage over multi-stream organization. Experiments on three Amazon benchmarks show that CC-MGR_{EC} consistently outperforms strong baselines in ranking quality while achieving more efficient full-graph representation refresh.

1 Introduction

Multimodal recommendation augments user-item interactions with visual and textual evidence that reveals item appearance, function, and usage context beyond item identifiers (He and McAuley 2016; Chen et al. 2017; Liu et al. 2025). Graph learning provides a natural way to exploit this evidence: node representations carry collaborative and multimodal information, while graph relations determine how that information is propagated.

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Existing graph-based approaches organize this information and its relations in several representative graph structures. MMGCN (Wei et al. 2019) maintains modality-specific propagation streams over the user-item interaction graph, whereas GRCN (Wei et al. 2020) uses multimodal signals to refine its interaction edges. LATTICE and MICRO (Zhang et al. 2021, 2023b) construct modality-aware item-item graphs from visual and textual similarity. FREEDOM (Zhou and Shen 2023) combines a denoised interaction graph with frozen item graphs, while MGCN (Yu et al. 2023) maintains separate behavior and content views. DOGE (Meng et al. 2025) further constructs an LLM-enhanced hyper-knowledge graph with item-item, user-user, and hyper-relations. Despite their different graph structures, these methods commonly assign modalities or relation types to separate propagation branches and fuse their resulting representations afterward.

This organization is more than an implementation choice. At each propagation layer, a node representation is updated using the information available to its branch, the relations traversable in its graph, the corresponding normalization, and its intermediate node representations. Separate branches therefore create distinct propagation contexts, an organization we characterize as the *context-decoupled paradigm*. It produces *recommendation-context fragmentation* at two levels. *Information fragmentation* prevents visual, textual, and collaborative information from jointly shaping the same intermediate node representation. *Relation fragmentation* prevents a collaborative-semantic bridge, formed by composing an interaction transition with a semantic transition, from arising within one message-passing path. Late fusion combines completed branch representations but cannot retroactively instantiate the cross-context updates omitted during their formation. The same organization also repeats sparse graph operations and maintains multiple representation streams. **But do the collaborative-semantic bridges and visual-textual compositions unavailable during context-decoupled propagation carry recommendation-relevant signal?**

Figure 1 tests this question at the two fragmentation levels. For relation fragmentation, Panel (a) uses the Baby training graph to compare whether collaborative-semantic bridges reach held-out positives more often than same-user unseen items with matched popularity and semantic reachability. The higher coverage among positives indicates that these bridges are associated with future user preference rather

(a) Relation Fragmentation (b) Evidence Fragmentation

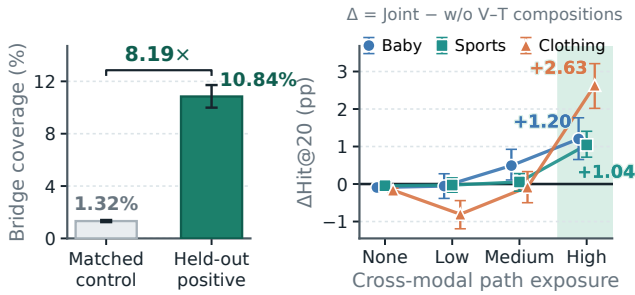


Figure 1: Motivation evidence for recommendation-context fragmentation. (a) Bridge coverage on Baby using only training interactions and the pre-calibration semantic graph; controls are matched in interaction degree, semantic support, and incoming semantic mass. (b) Ranking changes after removing paths that require both visual and textual transitions, with graph support, edge weights, normalization, initialization, first-order information, and all single-modality paths unchanged. Groups are defined by path exposure measured before removal; error bars are user-clustered 95% confidence intervals.

than basic graph connectivity. For information fragmentation, Panel (b) keeps the trained models fixed and removes only multi-step paths that require visual and textual transitions to interact, while retaining first-order information and every single-modality path. The effect concentrates on instances with greater exposure to the removed paths, indicating that these compositions contribute to prediction when they are available. Together, the two studies provide complementary evidence that context-decoupled propagation can isolate recommendation-relevant structure at both the relation and information levels.

Recovering the compositions lost to information and relation fragmentation is not achieved by simply merging branch outputs or graph edges. Content-induced connections may not reflect recommendation utility, and heterogeneous relations can compete when placed under shared propagation. We therefore propose **Context-Coupled Multimodal Graph Recommendation (CC-MGRec)** to couple these contexts reliably and efficiently. CC-MGRec first employs Recommendation-Grounded Semantic Calibration (RGSC) to make content-supported candidate relations more relevant to recommendation using cross-fitted training behavior, while uncertainty shrinkage and semantic-budget preservation stabilize the calibration. The calibrated relations are then combined with multimodal node information and observed interactions to form a Context-Coupled Multimodal Graph (CC-MG) under shared normalization. Finally, Collaborative-Semantic Compositional Propagation (CSCP) evolves a single representation stream over CC-MG, allowing heterogeneous information to jointly shape node representations and collaborative-semantic transitions to compose during propagation. Together, this design restores the information and relation compositions isolated by separate branches while avoiding their repeated graph passes and intermediate representations.

Our contributions are summarized as follows:

- We formulate propagation-context organization as a first-class problem in multimodal graph recommendation and identify recommendation-context fragmentation at the information and relation levels. Complementary structural and interventional diagnostics establish the recommendation relevance of the excluded compositions.
- We introduce the context-coupled paradigm and instantiate it with CC-MGRec through Recommendation-Grounded Semantic Calibration, construction of a Context-Coupled Multimodal Graph, and Collaborative-Semantic Compositional Propagation.
- We theoretically characterize the compositional coverage and computational advantage of context coupling over multi-stream propagation. Experiments on three Amazon benchmarks demonstrate consistent improvements in ranking quality and full-graph representation-refresh efficiency.

2 Related Work

Multimodal Recommendation

Multimodal recommendation enriches collaborative signals with visual, textual, acoustic, or multimodal-LLM evidence; a comprehensive taxonomy is provided by Liu et al. (2025). Early methods treat content as side information: VBPR models visual preference, while ACF attends to item- and component-level multimedia signals (He and McAuley 2016; Chen et al. 2017). Later methods learn multimodal representations through self-supervision or cross-view alignment, including SLMRec, BM3, and MMSSL (Tao et al. 2023; Zhou et al. 2023; Wei et al. 2023). Recent studies further improve evidence reliability through modality denoising, disentanglement, multimodal-LLM reasoning, semantic user initialization, or missing-modality completion (Ong and Khong 2025; Zhan et al. 2026; Dang et al. 2026; Xu et al. 2026; Li et al. 2026b). These studies mainly improve how multimodal evidence is extracted, aligned, fused, or initialized. Our work instead studies how multimodal evidence induces graph relations and how these relations should be organized with collaborative interactions during propagation.

Graph Learning for Recommendation

Graph-based recommenders differ in the information they use, the relations they construct, and how these relations participate in propagation. Table 1 summarizes these distinctions. An information source is marked when it contributes to recommendation learning, whereas a relation is marked only when it explicitly determines graph connectivity or message-passing support. CP indicates that collaborative and semantic relations jointly define one normalized propagation context, while SS indicates that only one evolving node-state stream is maintained. MMGCN propagates modality-specific representations over user-item graphs, GRCN refines interaction edges using multimodal evidence, and EgoGCN introduces edge-wise cross-modal modulation (Wei et al. 2019, 2020; Chen et al. 2022). These methods enrich interaction-graph learning but do not explicitly propagate content-induced item

Table 1: Comparison of representative graph recommenders. Tx/Im/In denote text, image, and interaction information; TR/VR/CR denote text-induced, visual-induced, and collaborative relations; CP and SS denote context-coupled propagation and a single state stream.

Method	Tx	Im	In	TR	VR	CR	CP	SS
MMGCN	✓	✓	✓	×	×	✓	×	×
GRCN	✓	✓	✓	×	×	✓	×	✓
LATTICE	✓	✓	✓	✓	✓	✓	×	×
MICRO	✓	✓	✓	✓	✓	✓	×	×
FREEDOM	✓	✓	✓	✓	✓	✓	×	×
MGCN	✓	✓	✓	✓	✓	✓	×	×
UGT	✓	✓	✓	×	×	✓	×	✓
SCALE	✓	✓	✓	✓	✓	✓	×	×
CC-MGREC	✓	✓	✓	✓	✓	✓	✓	✓

relations together with collaborative relations. LATTICE and MICRO construct modality-aware item graphs (Zhang et al. 2021, 2023b), while FREEDOM and MGCN combine collaborative and semantic information through separate graph views or propagation branches (Zhou and Shen 2023; Yu et al. 2023). Other methods introduce user co-occurrence, local-global structures, hyper-relations, signed relations, or stronger cross-view interaction (Wang et al. 2023; Guo et al. 2024; Hu et al. 2025; Meng et al. 2025; Xu et al. 2025; Dai et al. 2026; Lian et al. 2026). These designs enrich graph structure, but relation enrichment or representation fusion does not necessarily couple heterogeneous relations within the same propagation context. UGT unifies multimodal feature learning and aggregation on the interaction graph (Yi and Ounis 2024), whereas SCALE constructs a rich semantic graph but propagates it separately from the interaction graph (Li et al. 2026a). In contrast, CC-MGREC places observed user-item interactions and multimodal-induced item relations under one shared normalization and propagates one node-state stream over the resulting graph. This organization allows collaborative and semantic relations to jointly shape intermediate representations while avoiding repeated branch-wise propagation.

3 Preliminaries

Let $\mathcal{U} = \{u\}$ and $\mathcal{I} = \{i\}$ denote the user and item sets, with $M = |\mathcal{U}|$ and $N = |\mathcal{I}|$. Implicit feedback is represented by $\mathbf{R} \in \{0, 1\}^{M \times N}$, where $R_{ui} = 1$ indicates an observed interaction in the training set. We denote the interacted items of user u by $\mathcal{N}_u = \{i \mid R_{ui} = 1\}$, the users associated with item i by $\mathcal{U}_i = \{u \mid R_{ui} = 1\}$, and the item degree by $d_i = |\mathcal{U}_i|$.

Each item i is associated with a visual feature $\mathbf{x}_i^v \in \mathbb{R}^{d_v}$ and a textual feature $\mathbf{x}_i^t \in \mathbb{R}^{d_t}$ extracted by pretrained models. Our goal is to learn user and item representations $\mathbf{h}_u, \mathbf{h}_i \in \mathbb{R}^d$ and rank unobserved items according to

$$\hat{y}_{ui} = \mathbf{h}_u^\top \mathbf{h}_i. \quad (1)$$

We distinguish observed user-item interactions, denoted by $\mathcal{E}_c$, from content-supported item-item relations, denoted

by $\mathcal{E}_s$. Both relation families are defined over the common node set $\mathcal{V} = \mathcal{U} \cup \mathcal{I}$. For a directed weighted adjacency matrix $\mathbf{A}$, we adopt the destination-by-source convention: A_{pq} is the weight of the message sent from node q to node p . This convention allows calibrated item relations to carry different weights in opposite message directions.

4 Methodology

Framework Overview

Figure 2 presents the overall architecture of CC-MGREC. The model first transforms image and text content into stable semantic representations and uses interaction-derived item co-occurrence to identify how strongly each modality agrees with recommendation behavior. These signals define candidate item relations. Recommendation-Grounded Semantic Calibration (RGSC) then evaluates the candidate relations against cross-fitted training behavior and redistributes their semantic influence toward recommendation-relevant neighbors.

The calibrated item relations are coupled with multimodal node information and weighted observed interactions to construct a single Context-Coupled Multimodal Graph (CC-MG). All relation families share the same node space and normalization statistics. Collaborative-Semantic Compositional Propagation (CSCP) subsequently evolves one representation for every node over CC-MG, allowing multimodal information and heterogeneous relations to interact during graph learning rather than only after propagation.

Context-Coupled Multimodal Graph Construction

We first map the visual and textual features into a common d -dimensional space:

$$\mathbf{z}_i^m = \text{norm}(f_m(\mathbf{x}_i^m)), \quad m \in \{v, t\}, \quad (2)$$

where f_v and f_t are modality encoders and $\text{norm}(\mathbf{x}) = \mathbf{x}/\|\mathbf{x}\|_2$. The encoders are pretrained through modality reconstruction and symmetric image-text alignment, and are frozen before recommendation learning.

For modality m , the nonnegative content affinity between two items is $g_{ij}^m = [(\mathbf{z}_i^m)^\top \mathbf{z}_j^m]_+$. We retrieve the top- K neighbors of every item and symmetrize the resulting support:

$$\mathcal{E}_m = \{(i, j) \mid j \in \text{TopK}_{k \neq i}(g_{ik}^m) \vee i \in \text{TopK}_{k \neq j}(g_{jk}^m)\}. \quad (3)$$

Consequently, a relation discovered from either endpoint can support messages in both directions. The visual and textual supports remain distinguishable at this stage, allowing visual-only, text-only, and cross-modally supported item pairs to be retained.

Content similarity alone does not indicate which modality is most consistent with recommendation behavior. We therefore derive a collaborative reference from the training interactions. For two items i and j , their normalized user co-occurrence is

$$o_{ij} = \frac{|\mathcal{U}_i \cap \mathcal{U}_j|}{\sqrt{|\mathcal{U}_i| |\mathcal{U}_j|}}. \quad (4)$$

The top- K items according to o_{ij} form the collaborative reference neighborhood $\mathcal{C}_i$ of item i . This co-occurrence

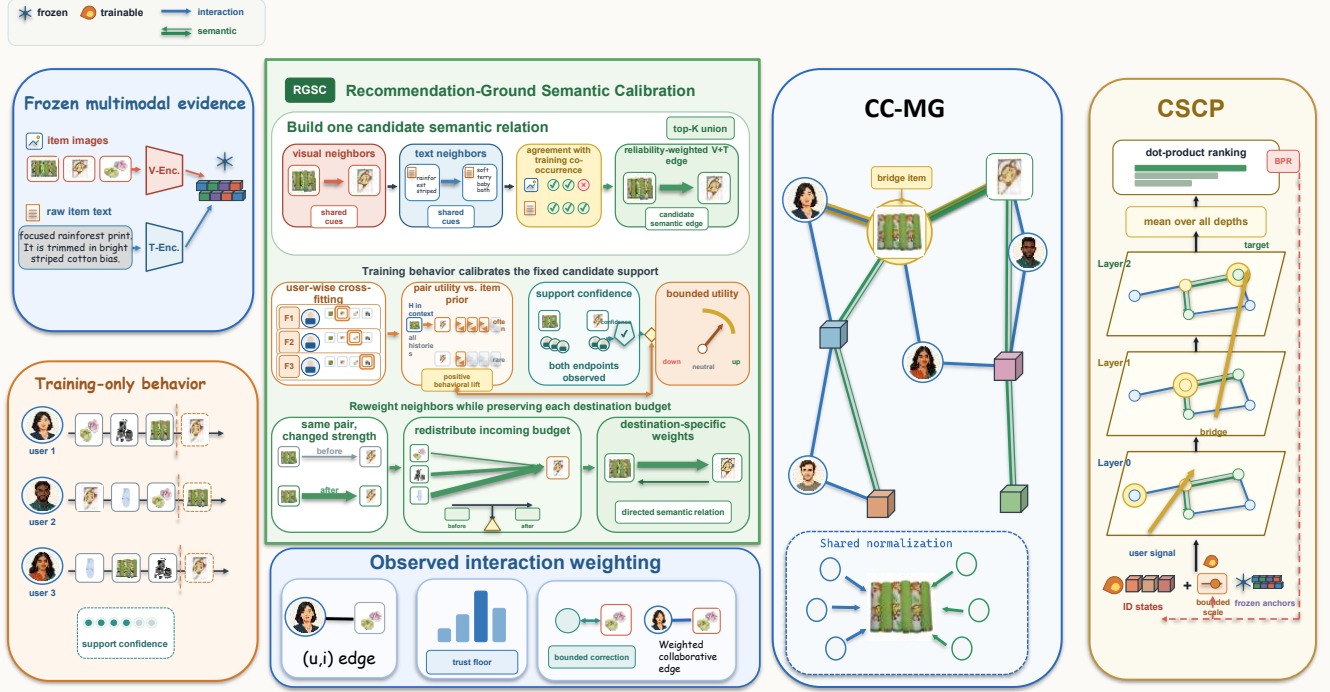


Figure 2: Overview of CC-MGREC. Multimodal item evidence and training-only collaborative evidence determine calibrated semantic relations, node representations, and observed-interaction weights. These signals are organized into one CC-MG under shared normalization. CSCP propagates a single representation stream over the resulting graph for recommendation.

structure is used to ground multimodal evidence; it does not introduce an additional propagation graph.

Let $\mathcal{N}_i^m$ denote the modality- m neighbors of item i . We measure the agreement between each modality and the collaborative reference by $a_m = (NK)^{-1} \sum_i |\mathcal{N}_i^m \cap \mathcal{C}_i|$. The global contribution of modality m is then

$$\alpha_m = \frac{\exp(a_m/\tau_\alpha)}{\sum_{m' \in \{v, t\}} \exp(a_{m'}/\tau_\alpha)}, \quad m \in \{v, t\}, \quad (5)$$

where τ_α controls the concentration of the modality weights and $\alpha_v + \alpha_t = 1$.

The union $\mathcal{E}_v \cup \mathcal{E}_t$ determines which item pairs may communicate. For every supported pair, we consolidate its available multimodal evidence into an initial semantic message weight:

$$w_{i \rightarrow j}^s = \beta \sum_{m \in \{v, t\}} \alpha_m g_{ij}^m \mathbb{I}[(i, j) \in \mathcal{E}_m], \quad (6)$$

where β controls the total strength of the semantic channel. Equation (6) retains evidence unique to either modality while accumulating agreement when both modalities support the same relation.

We next apply RGSC to determine how the semantic influence of each destination item should be distributed across its candidate neighbors. For every user with at least two training items, we partition the history into $F_u = \min(F, |\mathcal{N}_u|)$ folds. One fold serves as a target set and the remaining folds serve as context, with every fold taking the target role once.

For candidate direction $i \rightarrow j$, let $n_{i \rightarrow j}$ be the number of times that i occurs in a context while j occurs in the corresponding held-out fold. Let x_i be the total number of held-out target opportunities observed with context item i , and let p_j be the training-set popularity prior of item j . The recommendation lift of the candidate direction is

$$\ell_{i \rightarrow j} = \log \frac{(n_{i \rightarrow j} + \mu p_j)/(x_i + \mu)}{p_j}, \quad (7)$$

where μ controls prior smoothing. A positive lift indicates that i predicts j beyond the occurrence rate explained by the global popularity of j .

We combine the two directional lifts into $q_{ij} = \sigma((\ell_{i \rightarrow j} + \ell_{j \rightarrow i})/2)$. The support confidence $c_{ij} = \sqrt{x_i/(x_i + \mu) \cdot x_j/(x_j + \mu)}$ reduces the influence of relations whose behavioral evidence is uncertain. Their bounded recommendation-grounded reliability is

$$r_{ij} = c_{ij} \tanh\left(\frac{q_{ij} - 1/2}{\tau_r}\right), \quad (8)$$

where τ_r controls the sensitivity of the calibration.

Directly increasing reliable edges would also alter the overall semantic strength received by an item. To separate relation calibration from semantic-channel amplification, we preserve the original incoming semantic mass $M_j = \sum_k w_{k \rightarrow j}^s$ of every destination j :

$$\bar{w}_{i \rightarrow j}^s = M_j \frac{w_{i \rightarrow j}^s \exp(\gamma r_{ij})}{\sum_k w_{k \rightarrow j}^s \exp(\gamma r_{kj})}. \quad (9)$$

Here γ controls the calibration magnitude. The resulting weights satisfy $\sum_i \bar{w}_{i \rightarrow j}^s = M_j$, so RGSC changes which neighbors receive the semantic budget rather than increasing that budget. Because the conservation is applied separately at each destination, the calibrated semantic message graph can be directed even when its candidate support is symmetric.

The same multimodal evidence also supplies semantic information to the graph nodes. The frozen item anchor is

$$\mathbf{e}_i^s = \text{norm}(\alpha_v \mathbf{z}_i^v + \alpha_t \mathbf{z}_i^t). \quad (10)$$

Thus, visual and textual information jointly condition one item representation rather than producing independent downstream item states.

We obtain a training-only semantic profile for each user from the anchors of its interacted items. Each history item receives a normalized weight $\omega_{ui} \propto (d_i + 1)^{-1/2}$, preventing highly popular items from dominating the profile. Following the training-only semantic initialization principle of SG-URInit (Xu et al. 2026), we further blend a frozen cluster-level prototype $\mathbf{c}_u$ to stabilize short histories:

$$\mathbf{e}_u^s = \text{norm}\left(\sum_{i \in \mathcal{N}_u} \omega_{ui} \mathbf{e}_i^s + \xi \mathbf{c}_u\right), \quad (11)$$

where ξ controls the prototype correction.

Trainable ID embeddings preserve task-specific collaborative information, while the frozen semantic anchors inject multimodal evidence into the same node state:

$$\mathbf{h}_i^{(0)} = \mathbf{e}_i^{id} + s_i \mathbf{e}_i^s, \quad \mathbf{h}_u^{(0)} = \mathbf{e}_u^{id} + s_0 \mathbf{e}_u^s. \quad (12)$$

Here s_0 is a global semantic scale and $s_i = s_0 + \delta_i$ is a bounded, centered item-specific adjustment derived from frozen item evidence, its semantic neighborhood, and training degree.

Observed interactions define the collaborative support of CC-MG. We retain exactly the user–item pairs in $\mathbf{R}$ and assign the same weight to both message directions. For item i , the degree-aware trust is $\eta_i = \sigma(a \log(d_i + 1) + b)$. The collaborative edge weight is

$$C_{ui} = R_{ui} (\eta_i + (1 - \eta_i) [\cos(\mathbf{e}_u^s, \mathbf{e}_i^s)]_+). \quad (13)$$

The degree term preserves the reliability of well-supported observed behavior, while semantic consistency provides a bounded refinement for sparser interactions. Equation (13) reweights observed support without introducing unobserved user–item edges.

We now assemble all nodes and explicit propagation relations into CC-MG. Let $\bar{S}_{ji} = \bar{w}_{i \rightarrow j}^s$ and adopt the node order $\mathcal{U} \cup \mathcal{I}$. The weighted adjacency matrix is

$$\mathbf{A} = \begin{bmatrix} \mathbf{0} & \mathbf{C} \\ \mathbf{C}^\top & \bar{\mathbf{S}} \end{bmatrix} = \mathbf{A}_c + \mathbf{A}_s. \quad (14)$$

The zero block excludes user–user propagation. The off-diagonal blocks encode bidirectional observed interactions, whereas the item block contains calibrated semantic messages. Visual evidence, textual evidence, user histories, item co-occurrence, and observed interactions therefore affect graph learning through concrete node representations, relation support, and edge weights within the same topology.

Rather than normalizing collaborative and semantic relations independently, we compute one weighted incoming degree $D_{pp} = \sum_q A_{pq}$ over the complete graph. The shared normalized adjacency is

$$\hat{\mathbf{A}} = \mathbf{D}^{-1/2} \mathbf{A} \mathbf{D}^{-1/2}. \quad (15)$$

Consequently, the two relation families jointly determine both the messages received by a node and the scale at which those messages are propagated. No explicit self-loop is added; self-information is retained through the initial representation in the final layer aggregation. The semantic anchors and all graph weights are fixed before recommendation optimization.

Collaborative–Semantic Compositional Propagation

Given the context-coupled graph, CSCP performs lightweight propagation without relation-specific transformation matrices or branch-specific encoders. Let $\mathbf{H}^{(0)}$ stack all initial user and item representations. At layer l , the complete node state is updated by

$$\mathbf{H}^{(l+1)} = \hat{\mathbf{A}} \mathbf{H}^{(l)}, \quad l = 0, \dots, L - 1. \quad (16)$$

The same graph and the same representation stream are reused at every depth.

To expose how the two relation families meet inside this update, let $\mathcal{N}_c^-(p)$ and $\mathcal{N}_s^-(p)$ denote the collaborative and semantic in-neighbors of node p . Equation (16) is locally equivalent to

$$\mathbf{h}_p^{(l+1)} = \sum_{q \in \mathcal{N}_c^-(p)} \frac{w_{q \rightarrow p}^c}{\sqrt{D_{qq} D_{pp}}} \mathbf{h}_q^{(l)} + \sum_{q \in \mathcal{N}_s^-(p)} \frac{\bar{w}_{q \rightarrow p}^s}{\sqrt{D_{qq} D_{pp}}} \mathbf{h}_q^{(l)}. \quad (17)$$

For a user node, the semantic-neighbor set is empty. For an item node, collaborative messages from observed users and semantic messages from related items update the same representation. The multimodal information carried by the source states is therefore available to both relation families in subsequent layers.

For clarity, define $\mathbf{P}_c = \mathbf{D}^{-1/2} \mathbf{A}_c \mathbf{D}^{-1/2}$ and $\mathbf{P}_s = \mathbf{D}^{-1/2} \mathbf{A}_s \mathbf{D}^{-1/2}$. Consider an observed interaction $u \leftrightarrow i$ and a calibrated semantic relation $i \rightarrow j$. After two propagation steps, CSCP directly contributes

$$\mathbf{h}_j^{(l+2)} \supset (\mathbf{P}_s)_{ji} (\mathbf{P}_c)_{iu} \mathbf{h}_u^{(l)}, \quad \mathbf{h}_u^{(l+2)} \supset (\mathbf{P}_c)_{ui} (\mathbf{P}_s)_{ij} \mathbf{h}_j^{(l)}. \quad (18)$$

The first term transfers a user’s collaborative signal through an interacted item to a semantically related candidate. The second transfers the candidate’s semantic context back through the shared item to the user. Both computations reuse the intermediate representation of the same bridge item, which makes the collaborative–semantic composition part of graph learning itself.

We aggregate information from all propagation depths using

$$\mathbf{H} = \frac{1}{L+1} \sum_{l=0}^L \mathbf{H}^{(l)}. \quad (19)$$

Including $\mathbf{H}^{(0)}$ preserves each node’s trainable identity and frozen multimodal anchor while higher layers contribute increasingly composed graph evidence.

For an observed item i and a sampled unobserved item j of user u , we optimize Bayesian Personalized Ranking (Rendle et al. 2009):

$$\mathcal{L} = - \sum_{(u,i,j)} \log \sigma(\hat{y}_{ui} - \hat{y}_{uj}) + \lambda \|\Theta\|_2^2, \quad (20)$$

where Θ contains the trainable recommendation parameters. The multimodal encoders, semantic anchors, calibrated graph topology, and graph weights remain fixed during this stage.

5 Experiments

Research Questions

We organize the experiments around five questions: (RQ1) overall recommendation accuracy; (RQ2) recommendation-grounded calibration and graph construction; (RQ3) the contribution of mixed collaborative–semantic propagation; (RQ4) propagation efficiency and scalability; and (RQ5) robustness under user sparsity and key graph hyperparameters.

Experimental Setup

Datasets and protocol. We evaluate on three public Amazon categories widely used in multimodal recommendation: Baby, Sports and Outdoors, and Clothing, Shoes and Jewelry. We use the same filtered interaction data, visual/textual features, and train/validation/test protocol as the reproduced MIG-GT benchmark. For each test user, all items outside the masking set are ranking candidates. We report Recall and NDCG at cutoffs 10 and 20. Model selection uses validation Recall@20 only, and test results are read after selection.

Baselines. We compare with non-multimodal recommenders MF and LightGCN (Koren, Bell, and Volinsky 2009; He et al. 2020), adaptive or diffusion graph recommenders ApeGNN and MGDN (Zhang et al. 2023a; Hu et al. 2024), and multimodal methods VBPR, MMGCN, GRCN, DualGNN, and SLMRec (He and McAuley 2016; Wei et al. 2019, 2020; Wang et al. 2023; Tao et al. 2023). We further include item-graph methods LATTICE and FREEDOM (Zhang et al. 2021; Zhou and Shen 2023), the reproduced MIG-GT benchmark (Hu et al. 2025), and the recent DHMRec (Zhan et al. 2026). Baseline values follow their corresponding benchmark evaluations under the same Amazon categories; the strongest baseline is reproduced under the same data split and evaluation protocol.

Implementation details. Semantic and recommendation embeddings have dimension 128. The semantic encoder is pretrained and frozen before recommendation learning. Unless varied, visual and textual candidate neighborhoods use $K = 10$. The semantic strengths β are 1.5, 0.9, and 1.8, and propagation depths are 3, 2, and 2 for Baby, Sports, and Clothing, respectively. For RGSC, we use $F = 5$, $\mu = 10$, $\tau_r = 0.1$, and $\gamma = 1.0$ on all datasets; the user-wise fold assignment uses only training interactions and is held fixed within each paired comparison. We optimize BPR with batch size 8,000 and early stopping on validation Recall@20. All experiments run on one NVIDIA RTX 3090. The inherited item scale uses a stop-gradient predictor over ID state, frozen modality anchors, a frozen semantic-neighbor pool, and log

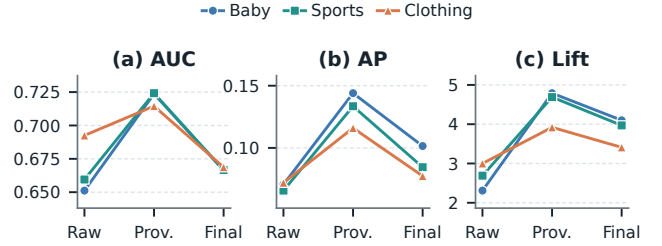


Figure 3: Held-out relevance of raw multimodal affinity and recommendation-grounded calibrated edge scores.

degree; its centered output is bounded by $|\delta_i| \leq \rho$ and is unchanged in all controlled comparisons. Unless otherwise specified, the main results in Table 2 are averaged over five random seeds, whereas all other experimental results are averaged over three random seeds. For efficiency evaluation, each run is further averaged over repeated CUDA-synchronized measurements after warm-up.

Overall Performance (RQ1)

Table 2 reports overall recommendation accuracy. CC-MGREC achieves the best result on every reported dataset-metric combination. Its advantage holds for Recall and NDCG at both cutoffs, showing that the improvement extends to both retrieval coverage and rank-sensitive quality rather than depending on one category or evaluation measure.

Recommendation-Grounded Calibration and Graph Construction (RQ2)

Table 4 separates semantics as graph topology from semantics as node initialization, and then removes recommendation-grounded utility calibration while keeping the candidate support, initialization, and trainable capacity fixed. Semantic topology yields substantial gains across all datasets, while grounding its edge weights in training behavior provides a further consistent improvement.

Figure 3 directly tests whether train-only calibration makes content-supported candidate relations more relevant to unseen interactions. Against the same fixed candidate support, the calibrated scores consistently improve AUC, average precision, and positive lift over raw multimodal affinity. This result complements the ranking ablation by showing that RGSC improves the recommendation relevance of the relations themselves.

Mixed-Relation Propagation (RQ3)

We remove only mixed collaborative–semantic products while preserving the graph, edge weights, shared normalization, initialization, capacity, and all relation-pure paths. Figure 4 shows that retaining these products improves both retrieval and ranking across datasets, isolating the recommendation value of cross-relation composition inside the joint operator.

6 Conclusion

We introduced CC-MGREC to study relation organization as a first-class design axis in multimodal graph recommendation.

Table 2: Overall performance on three datasets. R and N denote Recall and NDCG; the best baseline is underlined.

Method	Baby				Sports				Clothing			
	R@10	R@20	N@10	N@20	R@10	R@20	N@10	N@20	R@10	R@20	N@10	N@20
MF	.0357	.0575	.0192	.0249	.0432	.0653	.0241	.0298	.0206	.0303	.0114	.0138
LightGCN	.0479	.0754	.0257	.0328	.0569	.0864	.0311	.0387	.0361	.0544	.0197	.0243
ApeGNN	.0501	.0775	.0267	.0338	.0608	.0892	.0333	.0407	.0378	.0538	.0204	.0244
MGDN	.0495	.0783	.0272	.0346	.0614	.0932	.0340	.0422	.0362	.0551	.0199	.0247
VBPR	.0423	.0663	.0223	.0284	.0558	.0856	.0307	.0384	.0281	.0415	.0158	.0192
MMGCN	.0421	.0660	.0220	.0282	.0401	.0636	.0209	.0270	.0227	.0361	.0154	.0154
GRCN	.0532	.0824	.0282	.0358	.0599	.0919	.0330	.0413	.0421	.0657	.0224	.0284
DualGNN	.0513	.0803	.0278	.0352	.0588	.0899	.0324	.0404	.0452	.0675	.0242	.0298
SLMRec	.0521	.0772	.0289	.0354	.0663	.0990	.0365	.0450	.0442	.0659	.0241	.0296
LATTICE	.0547	.0850	.0292	.0370	.0620	.0953	.0335	.0421	.0492	.0733	.0268	.0330
FREEDOM	.0627	.0992	.0330	.0424	.0717	.1089	.0385	.0481	.0626	.0932	.0338	.0416
MIG-GT	<u>.0660</u>	.1015	<u>.0357</u>	<u>.0448</u>	<u>.0738</u>	.1126	<u>.0408</u>	<u>.0508</u>	.0628	.0926	<u>.0349</u>	<u>.0425</u>
DHMRc	.0646	<u>.1016</u>	.0343	.0436	<u>.0738</u>	<u>.1132</u>	.0401	.0502	<u>.0629</u>	.0931	.0340	.0417
CC-MGRc (Ours)	.0681	.1045	.0369	.0462	.0778	.1152	.0431	.0528	.0649	.0959	.0360	.0439

Table 3: Statistics of the evaluation datasets.

Dataset	Users	Items	Interactions
Baby	19,445	7,050	160,792
Sports	35,598	18,357	296,337
Clothing	39,387	23,033	278,677

Table 4: Ablation of semantic topology and recommendation-grounded utility calibration (R@20/N@20).

Variant	Baby	Sports	Clothing
Full weighted CC-MG	.1064/.0467	.1155/.0527	.0980/.0445
w/o semantic topology	.0956/.0423	.1057/.0479	.0726/.0330
w/o utility calibration	.1055/.0459	.1133/.0516	.0922/.0416

RGSC constructs a mass-conserving, recommendation-aware CC-MG from multimodal item relations and observed interactions, and CSCP propagates this graph through one shared operator. The resulting powers are closed over valid collaborative-semantic relation sequences, directly exposing mixed walks that separated branches do not instantiate during propagation. Consistent gains under relation removal, path removal, matched separation, sparsity analysis, and synchronized latency measurement support the central conclusion: organizing calibrated relations in one propagation space can improve both recommendation quality and representation-refresh efficiency without adding model capacity.

Reproducibility Statement

The final release will include preprocessing scripts, train/validation/test splits, all hyperparameter search ranges, random seeds, hardware and software information, and commands for every main and auxiliary experiment. All checkpoints will be selected using validation data only, and the test

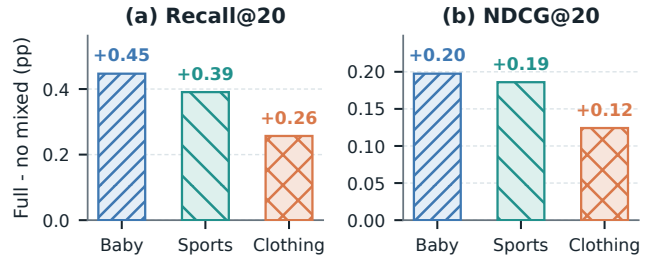


Figure 4: Recommendation gain of full CC-MG over the matched no-mixed intervention.

set will be evaluated after model selection.

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